

Mesa's Intel ANV Vulkan driver gets AV1 encoding

Mesa's open source Intel ANV Vulkan driver has gained hardware-accelerated AV1 encoding through the VK_KHR_video_encode_av1 extension, exposing the dedicated AV1 encoding hardware in Intel's DG2/Alchemist GPUs, including Arc A-Series graphics, to Linux applications through a standardised API.

The update does not add AV1 encoding to the GPUs; instead, it provides Vulkan applications with an open interface to hardware already present in the chips. This strengthens Mesa and Vulkan Video as an alternative to vendor-specific video acceleration APIs.

Finalised by the Khronos Group in 2024 and included in Vulkan 1.3.302, the AV1 encoding extension complements existing AV1 decoding and enables applications to query capabilities, configure encoding sessions, control rate settings, manage reference frames, and provide AV1-specific parameters.

With AV1 joining H.264 and H.265, Vulkan Video can now offer standardised interfaces for encoding and decoding all three codecs. The development follows July 2026 work that re-enabled H.264 and H.265 encoding for Intel Gen12.5/Alchemist hardware in ANV after insufficient testing had led to their earlier disablement.

The capability could benefit open source multimedia projects such as FFmpeg and GStreamer, which are working on Vulkan Video support. However, applications must implement the functionality before users can access it.



shown in users' feeds. X has also allowed external researchers studying recommendation systems to train the Phoenix scoring system using the open source code.

Along with the code release, X is offering users features that allow them to analyse how its ranking system has affected their accounts or posts. The 'Under the Hood' transparency feature will be available through the app's settings. Users who have posted at least 10 times during the previous month can download aggregated statistics in JSON format showing whether their account or post received any labels during the previous calendar month.

The transparency initiative also includes access to the source code used for recommendations. However, some parts of the system are not being made public, including systems that use the Grok machine learning model to predict whether a post may violate a rule. X says these details are being withheld to prevent malicious users from exploiting the system.

Zettabyte launches MaaS, a managed cloud service

Zettabyte has launched Model-as-a-Service (MaaS) on zCLOUD, turning access to open source and open-weight AI models into an on-demand cloud service through



a single API. The service gives organisations access to models including Kimi, GLM, Llama, GPT-OSS, Gemma, and Qwen without requiring them to

independently deploy and manage each model.

The models run on Zettabyte's Nvidia H100, H200, B200, and B300 GPUs. Teams can call them through APIs without sourcing, installing, or tuning the models themselves. Organisations requiring greater control can instead rent raw GPU capacity on zCLOUD and run their own models.

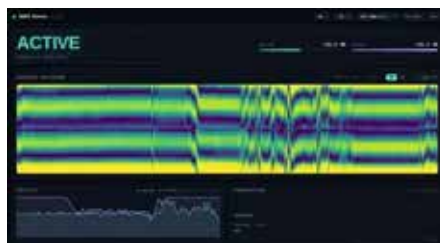
Nvidia H100 capacity is available immediately, starting at US\$1.99 per GPU-hour depending on requirements and scale. zCLOUD targets 99.5% uptime, while reserved clusters are designed for deployment within six hours of confirmed requirements.

Zettabyte offers short-term GPU capacity, reserved clusters for sustained training and inference, and private cloud deployments for procurement, security, or regional requirements.

"zCLOUD was built to deliver Model-as-a-Service, bringing more than 40 hardware providers together and managing the underlying AI infrastructure through our zSUITE full stack," said Dr David Ku, technology leader at Zettabyte.

WifiSense-Pi project detects human presence through walls

An open source project uses Wi-Fi signal disturbances to detect subtle movement



and breathing, allowing it to identify human presence through walls.

The WifiSense-Pi project uses an ESP32-S3 and Raspberry Pi 4 to detect human presence through changes in Wi-Fi signals. The ESP32-S3 measures the radio channel 100 times per second,

while the Raspberry Pi 4 handles signal processing. The system can detect subtle movements and identify a stationary person through changes caused by breathing.

Instead of transmitting dedicated sensing signals, the project analyses existing signals in the 2.4GHz Wi-Fi band. Human presence and movement affect these signals, allowing the system to detect changes without traditional motion sensors. The technique can also detect presence through walls, which is an advantage over conventional motion sensors.